Luis Maximiliano Labán Seminario

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Profile

Physician and epidemiologist working on burden of disease, health-data analysis, and responsible use of AI in health. Completed the coursework of the MSc in Epidemiological Research (UPCH) and is an MSc candidate in Statistical Engineering (UNI). Experience at CDC Peru (Ministry of Health), IETSI-EsSalud and, in 2026, estimating years lived with disability for Peru's national burden-of-disease study. Builds reproducible R, Python, and SQL workflows; co-author of seven peer-reviewed articles; RENACYT-registered researcher.

Education

MSc Candidate, Statistical Engineering (Maestría en Ciencias e Ingeniería Estadística). Universidad Nacional de Ingeniería (UNI), Lima, Peru. *2025–present.*

MSc in Epidemiological Research, coursework completed (57 credits). Universidad Peruana Cayetano Heredia (UPCH), Lima, Peru. *2023.*
Thesis in progress: high-sensitivity C-reactive protein and metabolic syndrome in the CRONICAS population cohort (coast and high-altitude Peru). Advisor: Antonio Bernabé-Ortiz. Degree to be awarded upon thesis defence.

Medical Doctor (Médico Cirujano). Universidad Nacional de Piura, Piura, Peru. *2014–2021.*

Professional Experience

Individual Consultant, Burden of Disease. PCRIS Programme, Ministry of Health, Peru. *Aug–Dec 2026.*

- Data analysis to estimate years lived with disability (YLDs) for Peru's 2024 burden-of-disease study: parameter matrices, reproducible R scripts, and technical reports.

Service Contractor, Clinical-Guideline Chatbot Project. Faculty of Human Medicine, Universidad de San Martín de Porres (USMP), Lima, Peru. *Oct–Nov 2026.*

- Professional services for the development and validation of an expert chatbot on clinical practice guidelines: quality control and methodological documentation.

Teaching Assistant (Jefe de Práctica), Epidemiology. School of Public Health and Administration, UPCH, Lima, Peru. *Aug–Nov 2025.*

- Practical sessions of the Epidemiology course, MSc in Epidemiological Research: epidemiological methods, data analysis, and interpretation of results.

Statistical Analyst; Scientific Copy Editor. Faculty of Human Medicine, USMP, Lima, Peru. *May–Jun 2025.*

- Statistical analyst for a community-based project on premalignant cervical lesions (Chavinillo, Huánuco): study design, database structure, analysis plan, analysis, and interpretation.
- Scientific copy editing of the final research report for methodological, terminological, and editorial coherence.

Service Contractor, Clinical Epidemiology and Health-Systems Studies. Institute for Health Technology Assessment and Research (IETSI), EsSalud, Lima, Peru. *Jan–Apr 2025.*

- Designed statistical methods, including survival analyses and health-programme impact evaluation, for studies in cancer, renal health, and other chronic non-communicable diseases.
- Wrote the protocol and extracted, cleaned, and analysed data for the RRI 01-2025 report on national chronic-kidney-disease screening (VISARE programme, 2013–2022).
- Methodological advice for studies linked to the Kaelin Prize; automated data validation, cleaning,

and reproducible analysis in R and Python.

Epidemiological and Public-Health Data Analyst. National Centre for Epidemiology, Prevention and Disease Control (CDC Peru), Ministry of Health, Lima, Peru. *Apr–Dec 2024.*

- Extracted, cleaned, integrated, validated, and analysed large healthcare databases with R and SQL, with scripts, technical documentation, and reproducible workflows.
- Analysed morbidity, mortality, burden-of-disease, and public-health impact indicators; prepared technical reports.
- Wrote a methodological manual for estimating public-health indicators and a monitoring-and-evaluation guide; synthesised evidence on community and government programmes to reduce anaemia in Peru.

General Physician, Haemodialysis. Centro de Diálisis Piura, Piura, Peru. *Oct 2022–Dec 2023.*

- Care of patients with chronic kidney disease before, during, and after haemodialysis; management of procedure-related complications; laboratory follow-up with a multidisciplinary team.

General Physician, Rural and Urban-Marginal Health Service (SERUMS). Health Facility I-2 Parachique, Regional Health Directorate Piura, Sechura, Peru. *Oct 2021–Sep 2022.*

- Primary care, prevention, and health promotion; epidemiological surveillance, immunisation campaigns, and health-information reporting.

Research Outputs

Registered protocol and technical report

- **First author.** Labán-Seminario LM, Soto-Becerra P, De León Delgado J, Sánchez Villar LA, Salinas Paz JL. *Overall survival and quality of life in cancer patients treated with preparations of extracts of Viscum album L (European mistletoe): protocol of a systematic review and meta-analysis.* PROSPERO. 2026; CRD420261336349.
- *RRI 01-2025: National chronic-kidney-disease screening during 2013–2022, based on VISARE programme data.* EsSalud technical report, 2025. hdl:20.500.12959/5686.

Peer-reviewed journal articles

1. Bendezu-Quispe G, **Labán-Seminario LM**, Arce-Huamani MA, et al. Biomedical informatics: characterization of the offer of massive open online courses. *Medwave.* 2022. doi:10.5867/medwave.2022.11.2631.
2. **Labán-Seminario LM**, Carrillo-Larco RM, Bernabé-Ortiz A. Stroke-related length of hospitalization trends and in-hospital mortality in Peru. *PeerJ.* 2022;10:e14467. doi:10.7717/peerj.14467.
3. Porras M, Diez-Canseco F, Paredes M, **Labán-Seminario LM**, Zafra-Tanaka JH, Praveen D, Miranda JJ. Going digital: opportunities and barriers in the use of technology for health research. *Salud Pública de México.* 2022;64:S22–S30. doi:10.21149/12977.
4. Calderon-Ticona JR, Taype-Rondan A, Villamonte G, **Labán-Seminario LM**, Helguero-Santín LM, Miranda JJ, Lazo-Porras M. Diabetes care quality according to facility setting: a cross-sectional analysis in six Peruvian regions. *Primary Care Diabetes.* 2021;15(3):488–494. doi:10.1016/j.pcd.2020.11.014.
5. Mezones-Holguin E, Gamboa-Cardenas RV, Sanchez-Felix G, et al., **Labán-Seminario LM**, et al. Efficacy and safety in the continued treatment with a biosimilar drug in patients receiving infliximab: a systematic review in the context of decision-making from a Latin-American country. *Front Pharmacol.* 2019;10:1010. doi:10.3389/fphar.2019.01010.
6. Culquichicon C, Helguero-Santin LM, **Labán-Seminario LM**, Cardona-Ospina JA, Aboshady OA, Correa R. Massive open online courses in health sciences from Latin American institutions: a need for improvement? *F1000Research.* 2017;6:940. doi:10.12688/f1000research.11626.1.
7. Culquichicon C, Hernandez-Pacherres A, **Labán-Seminario LM**, Cardona-Ospina JA, Rodriguez-Morales AJ. Where are we after 60 years of paragonimiasis research? A bibliometric

assessment. *Infez Med.* 2017;25(2):142–149.

Conference abstracts

- Calderon-Ticona JR, et al., **Labán-Seminario LM**, Miranda JJ. Coverage of diabetes care according to facility level in Peru. International Diabetes Federation (IDF) Congress, Abu Dhabi, December 2017.
- Culquichícon C, et al., **Labán-Seminario LM**, et al. Massive open online courses in health sciences from Latin American institutions. TASME Spring Conference, Keele, UK, April 2017.

Full bibliography: NCBI MyBibliography.

Research Funding and Fellowships

GEOHealth Fellowship, Global Environmental and Occupational Health (2018). Fogarty/NIH GeoHealth Hub at UPCH, Lima; project 5U01TW010107-04 (NIH RePORTER).

Research grant, Universidad Nacional de Piura (2017). Resolution 1283-2017-OPPTO-OCP-UNP; total budget PEN 23,000.

Recognition as speaker, "Semilleros de investigación", Universidad Nacional de Piura (2017).

Open-Source Projects (2026, in development)

- **ENDES.metodos.** R package for complex-survey analysis, public-health indicators, and spatial small-area estimation with Peruvian population surveys (experimental). github.com/llabans/ENDES.metodos.
- **grafos-LLM.** Clinical practice guidelines transformed into knowledge graphs, with data provenance, leakage prevention, and human review; foundation for a guideline chatbot (not a clinically validated model). github.com/llabans/grafos-LLM.
- **Breast-cancer navigation-programme evaluation (USMP).** Record linkage, cohort reconstruction, and computable phenotypes from pseudonymised oncology records.
- **scienceapp.** PubMed evidence-synthesis module of a research-support platform. github.com/llabans/scienceapp.

Teaching

Teaching Assistant, Epidemiology, MSc in Epidemiological Research, UPCH (Aug–Nov 2025).

Workshop facilitator, Faculty of Medicine, Universidad Nacional Mayor de San Marcos (UNMSM), 2025: statistics and data analysis; AI in teaching and research; scientific writing and publication; research-project development; literature-search methods (five workshops, Jul–Nov 2025).

Selected Training

- Global burden of disease: evidence and tools for public-health management. UPCH, 2026 (32 h; grade 19.55/20). Earlier: The global burden of disease: metrics for evaluation in public health, UPCH, 2018.
- Target validation in the era of genomics, big data, and AI. Wellcome Sanger Institute (virtual), Sep 2026.
- AI competencies for research: evidence synthesis (UPCH, 2026); Higher-education teaching (UPCH, 2026).
- Six AWS Training & Certification courses on ML, generative AI, foundation models, and responsible AI (Jan 2026); Supervised Machine Learning: Regression and Classification (Stanford Online/DeepLearning.AI, 2024).
- Outbreak analysis and public-health modelling (UPCH, 2024); Introduction to causal inference in health (USIL, 2018); III International Course on Environmental Health (GEOHealth: Emory, UPCH, U. de Chile, Johns Hopkins, 2018).
- Research ethics: CITI Responsible Conduct of Research (2026, 2023); CITI Biomedical Research (2023); Protection of Human Research Subjects (CITI/NIH, 2017).

Skills, Registration, and Languages

Programming. R (advanced); Python (intermediate); Stata (intermediate); SQL (basic–intermediate); Julia (basic). Quarto, Git, Docker (basic), RStudio, VS Code, Jupyter.

Methods. Survival analysis; health-programme evaluation; morbidity, mortality, and burden-of-disease indicators (incl. YLDs); complex surveys and small-area estimation; evidence synthesis; reproducible data pipelines.

Registration. Peruvian Medical Council (CMP 094109, active); RENACYT researcher, Level VI (P0072109).

Languages. Spanish (native); English (intermediate–advanced); Chinese (basic).